RECOVERY LIVES IN U, CONDITIONING ACTS IN R: WHERE ARBITRARY-SCALE SUPER-RESOLUTION GAINS COME FROM

Anonymous authors

Draft for internal review

ABSTRACT

Arbitrary-scale super-resolution (ASISR) networks pair an image encoder with an implicit decoder that also receives the target pixel size (cell). Decoder and cell designs have multiplied, yet their gains shrink at large scales. We explain this with two frequency coordinates. Measured per band, recovery is a function of frequency normalised by the low-resolution Nyquist limit, u. It is the same for four decoder designs on a shared encoder (within 0.01). Training and a stronger encoder raise it only for $u \geq 1$. The cell, in contrast, acts on the output as a low-pass in high-resolution frequency r. For large mismatch, its cost is predicted by the recovered energy at the frequencies it attenuates. Because a fixed r moves to larger u as scale grows, where little is recovered, the cell's leverage vanishes. Doubling the cell costs 2.1–2.5 dB at $\times 2$, 0.13 dB at $\times 4$ and at most 0.03 dB at $\times 6$. All tests were pre-registered with frozen predictions; failed predictions are reported. Matched training confirms that a stronger encoder (RDN, SwinIR) raises recovery only above the LR Nyquist limit (+0.17 to +0.31 dB) and that removing the cell costs 0.09 dB at $\times 2$ but under 0.01 dB at $\times 6$. Decoder *design* barely matters, whereas cutting decoder *capacity* costs 0.14–0.20 dB at every scale, which contradicts our frozen prediction.

***Index Terms*—** Arbitrary-scale super-resolution, implicit neural representation, frequency analysis, scale conditioning

1. INTRODUCTION

LIIF [1] reconstructs an image at continuous coordinates. It decodes local encoder features together with the query offset and the target cell size. Many decoders followed: LTE's local Fourier estimator [2], integrated positional encoding [3], attention-based decoders [4], neural operators [5], Gaussian splatting [6] and filtered heat fields [7]. A recent benchmark finds that gains from decoder design are small and saturate [8]. Frequency analyses describe low-pass behaviour [9], probe responses [10] or scale anchoring [11]. They do not say *where* a decoder or the cell can still change the output, or why that space disappears as the scale grows.

We answer with two coordinates. For scale s and HR frequency r (cycles per HR pixel), let $u = 2sr$ be the frequency relative to the LR Nyquist limit. Our claims are:

1. **Recovery lives in u.** Per-band recovery $R(u)$ is nearly scale-independent and decoder-independent. Training and the encoder change it only above $u \approx 0.75$.

2. **Conditioning acts in r.** Changing the cell multiplies the native output by an approximately scale-independent low-pass $w(r)$. Its cost is bounded by the recovered energy at those r, so the cell's leverage decays with scale.

3. **Consequence.** The cell input can be dropped at large scales, and decoder design is not a lever. The capacity that learns the above-Nyquist prior, whether in the encoder or in the decoder MLP, is. We test this with matched training.

That cell inputs can occasionally hurt is already known [3]. Our contribution is a quantitative account of where any decoder- or cell-side change can act at all. Every experiment followed a frozen protocol (code, predictions and scoring hashed before results); Table 2 lists outcomes, including failures.

2. SETUP

Models. LIIF, LTE and SC-INR (an LTE-derived variant whose cell enters through an analytic sinc weight) share an EDSR-baseline encoder [12] and the LIIF training recipe (DIV2K [13], scales $\mathcal{U}(1, 4)$, 1000 epochs, three seeds). We add GaussianSR [6] (EDSR-baseline encoder) and RDN-BTC [14, 15]. For matched comparisons we train LIIF for 200 epochs with an identical recipe and EDSR-baseline, RDN or SwinIR [16] encoders, a reduced decoder (2×64 MLP) or no cell input.

Data and degradation. Fresh DIV2K valid 0869–0900 (never used in this project), Set14 [17], BSD100 [18] and Urban100 [19]. We take centre 240^2 crops and bicubic anti-aliased downsampling at $\times\{2, 3, 4, 6\}$; $\times 6$ lies outside training support.

Measurements. After a 4-pixel shave, errors are split by Chebyshev FFT radius into u-bins of width 0.25. Then $R(u) = 1 - \mathrm{MSE}_u / E_u$, with E_u the target energy in the bin. The cell sweep multiplies the native cell by $m \in [\frac{1}{2}, 2]$ (11 values).

3. RECOVERY LIVES IN U

Fig. 1a shows $R(u)$ for four scales. Below the LR Nyquist limit ($u < 1$) networks and bicubic interpolation both recover most energy. Between $u = 0.75$ and 2 the network recovers far more: at $u = 1$, 0.68 versus 0.20 for bicubic; at $u = 1.5$, 0.36 versus 0.04. Curves at different scales nearly coincide, but only approximately. Our frozen tolerance (spread ≤ 0.05) held in 45/54 cases; the misses sit in the $\times 2$ bin closest to the HR Nyquist limit.

Decoders do not move $R(u)$. On a shared EDSR-baseline encoder, LIIF, LTE, SC-INR and GaussianSR differ by at most 0.010 in every bin (18/18 and 18/18; Fig. 1b). Correspondingly, their PSNR differs by only 0.03–0.06 dB, shrinking from $\times 2$ to $\times 6$.

Training and the encoder do. Between epoch 100 and 1000, $R(u)$ rises by up to 0.045 for $u \geq 1$ (12/12 bins). It is unchanged below $u < 0.75$ (≤ 0.008; 9/9; Fig. 1c). With decoder and recipe fixed, an RDN encoder raises $R(u)$ by up to 0.053 for $u \geq 1$ (12/12) and leaves $u < 0.75$ unchanged (9/9). A SwinIR encoder does the same at an equal 50 epochs (12/12, 9/9; Fig. 1c). External checkpoints (RDN-BTC, official EDSR) underperformed under our degradation and are not used as encoder evidence.

4. CONDITIONING ACTS IN R

Let P_1 be the native output and P_m the output with cell multiplied by m. Projected per band, we find

$$P_{m,b} \approx w_m(r_b)\, P_{1,b}, \quad w_m(r) \in \mathbb{R}, \tag{1}$$

a low-pass in HR frequency whose shape barely depends on scale (Fig. 2a). Writing $e_b = T_b - P_{1,b}$ for the native error, Eq. (1) gives the exact expansion

$$\Delta\mathrm{MSE}_b = 2(1-w)\,\mathrm{Re}\langle e_b, P_{1,b}\rangle + (1-w)^2\|P_{1,b}\|^2. \tag{2}$$

If the native cell is calibrated, the first-order term is small. The cost is then governed by $(1-w)^2\|P_{1,b}\|^2$: the energy the network actually emits at the frequencies the cell attenuates. At fixed r, larger scales place that energy at larger u, where little has been recovered. This is why the cell's leverage decays with scale.

Tests. For $m = 2$, the cost peaks at $r \approx 0.25$ for every model, seed and scale (36/36; Fig. 2b). Its magnitude tracks recovered energy (Spearman 0.77–0.78). The second-order term ranks per-band cost with $\rho = 0.86$–0.91 and accounts for 70–77% of it. Cost decays monotonically with scale for all nine model–seed pairs (Fig. 2c), and also for GaussianSR and RDN-BTC. The native cell is close to calibrated: the best fixed multiplier improves on it by at most 0.017 dB. Its optimum drifts slightly above $m = 1$ at $\times 4$–$\times 6$, so the frozen "native is optimal" gate failed (20/27).

Limits of the model. For small mismatch ($m = 4/3$) the first-order term dominates and Eq. (1) does not predict cost.

Table 1. Matched LIIF training (200 epochs, seed 1): PSNR change (dB, non-DC) relative to EDSR-baseline LIIF on DIV2K 0869–0900. †SwinIR compared with EDSR at an equal 50 epochs.

Arm	$\times 2$	$\times 3$	$\times 4$	$\times 6$
decoder 2×64	-0.20	-0.17	-0.15	-0.14
no cell input	-0.09	-0.03	-0.04	-0.00
RDN encoder	$+0.31$	$+0.23$	$+0.23$	$+0.17$
SwinIR encoder†	$+0.17$	$+0.18$	$+0.12$	$+0.07$

The peak at $r \approx 0.25$ holds only for the LIIF family; GaussianSR and BTC peak elsewhere. We had also predicted that a too-small cell (requesting detail) would cost more than a too-large one, by analogy with prior-mismatch asymmetry. The opposite held (33/144), so we withdraw that hypothesis.

5. CONSEQUENCE: WHAT IS A LEVER

Before training we froze three design predictions (Table 1). (D2) Removing the cell input costs little at large scales. This largely held: 0.09 dB at $\times 2$, 0.03–0.04 dB at $\times 3$/$\times 4$ and 0.004 dB at $\times 6$. The gate still failed because $\times 4$ slightly exceeded $\times 3$ and the 0.03 dB bound. (D3) A stronger encoder helps more than a reduced decoder hurts. This held: RDN adds 0.23 dB at $\times 4$, and both RDN and SwinIR raise $R(u)$ only above the LR Nyquist limit. (D1) A $4\times$ smaller decoder MLP costs ≤ 0.05 dB at $\times 4$/$\times 6$. This *failed*: it costs 0.14–0.20 dB at every scale, again only for $u \geq 0.75$. The decoder MLP therefore also stores part of the above-Nyquist prior. The correct reading is that decoder *design* is not a lever (four designs, identical $R(u)$), while learned capacity anywhere in the network is. Per parameter, the decoder MLP is the more efficient place for it (about 0.3M parameters for 0.15 dB, against about 20M for RDN's 0.23 dB). The scale-dependent cell is the part that can be dropped at large scales.

6. LIMITATIONS AND CONCLUSION

We study bicubic degradation, centre crops, convolutional and Transformer encoders and LIIF-style training. The u-collapse is approximate. The filter model of the cell holds for large mismatch only. Its frequency locus is specific to decoder families. Within these limits, ASISR gains lie in what the network learns above the LR Nyquist limit. Matched arms use one seed and 200 (SwinIR 50) epochs. The cell is an HR-frequency control whose reach is set by that recovery. Decoder and cell innovations should therefore be evaluated where $u \lesssim 2$ at the scale of interest, while the scale-dependent cell can be dropped at large scales. Learned capacity, not decoder form, sets the above-Nyquist prior.

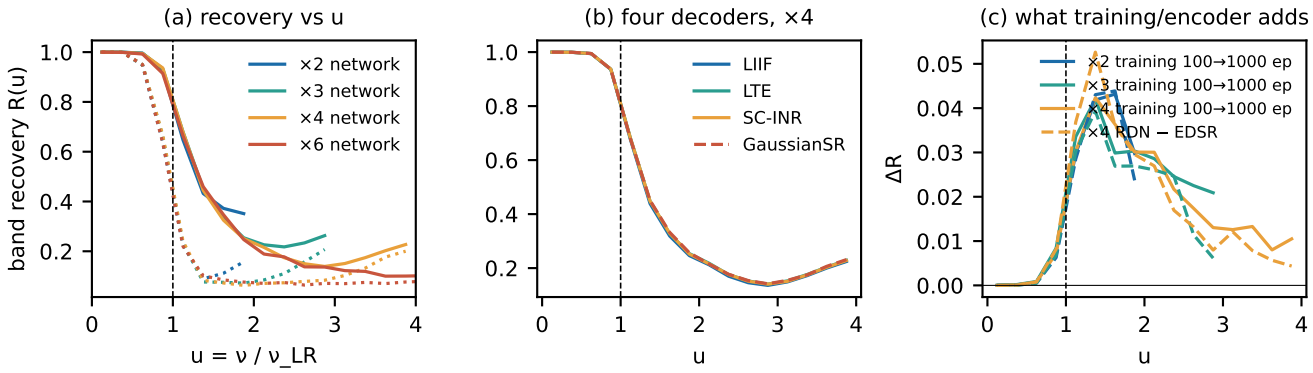


Fig. 1. (a) Band recovery $R(u)$ at four scales (solid: mean of three decoders and three seeds; dotted: bicubic). (b) Four decoders on the same encoder. (c) Change in $R(u)$ from further training and from a stronger encoder.

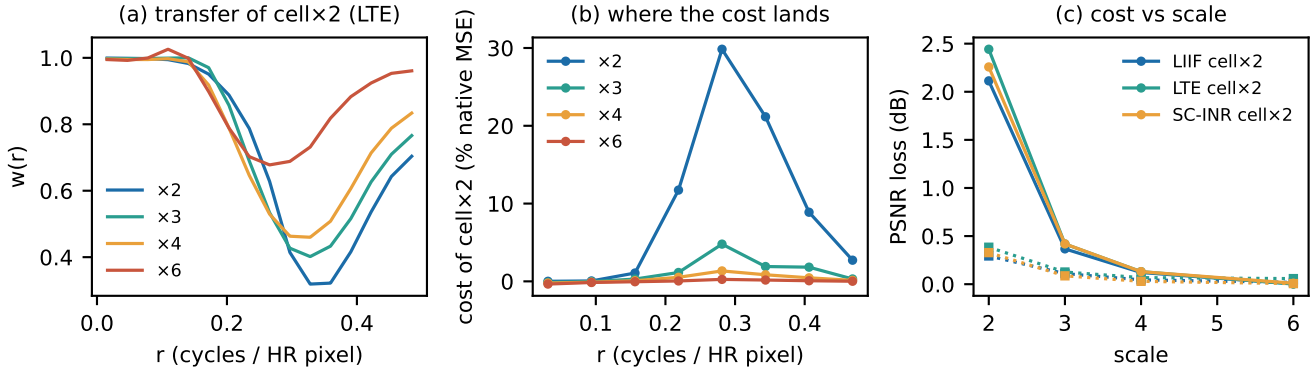


Fig. 2. (a) Spectral transfer $w(r)$ of doubling the cell (LTE, Set14). (b) Where its cost lands in HR frequency. (c) PSNR cost of doubling (solid) or halving (dotted) the cell, by scale.

7. REFERENCES

[1] Yinbo Chen, Sifei Liu, and Xiaolong Wang, "Learning continuous image representation with local implicit image function," in *CVPR*, 2021.

[2] Jaewon Lee and Kyong Hwan Jin, "Local texture estimator for implicit representation function," in *CVPR*, 2022.

[3] Ying-Tian Liu, Yuan-Chen Guo, and Song-Hai Zhang, "Enhancing multi-scale implicit learning in image super-resolution with integrated positional encoding," arXiv:2112.05756, 2021.

[4] Jiezhang Cao et al., "CiaoSR: Continuous implicit attention-in-attention network for arbitrary-scale image super-resolution," in *CVPR*, 2023.

[5] Min Wei and Xuesong Zhang, "Super-resolution neural operator," in *CVPR*, 2023.

[6] Jintong Hu, Bin Xia, Bin Chen, Wenming Yang, and Lei Zhang, "GaussianSR: High fidelity 2d gaussian splatting for arbitrary-scale image super-resolution," in *AAAI*, 2025.

[7] Alexander Becker, Rodrigo Caye Daudt, Dominik Narnhofer, Torben Peters, Nando Metzger, Jan Dirk Wegner, and Konrad Schindler, "Thera: Aliasing-free arbitrary-scale super-resolution with neural heat fields," *Transactions on Machine Learning Research*, 2025.

[8] Tayyab Nasir, Daochang Liu, and Ajmal Mian, "Implicit neural representation-based continuous single image super-resolution: An empirical benchmark," arXiv:2601.17723v3, 2026.

[9] Haoyu Deng, Zijing Xu, Yule Duan, Xiao Wu, Wenjie Shu, and Liang-Jian Deng, "Exploring the low-pass filtering behavior in image super-resolution," in *ICML*, 2024.

[10] Shusei Kawai, Fumiya Arai, Takao Marukame, Tetsuya Asai, Alexandre Schmid, and Kota Ando, "Frequency-

Table 2. Pre-registered predictions and outcomes.

Prediction	Outcome
$R(u)$ spread across scales ≤ 0.05	45/54 **fail**
decoders within 0.03 (four decoders)	18/18, 18/18
training raises only $u \geq 1$	12/12; 9/9
cell$\times 2$ cost peaks at $r \in [0.19, 0.375)$	36/36 (LIIF family)
cost $\sim$ recovered energy ($\rho \geq 0.7$)	0.77–0.78
leverage decays with scale	9/9
native cell optimal	20/27 **fail**
too-small cell costs more	33/144 **fail**
$\geq$80% of cost at $u \geq 1$	75% **fail**
RDN raises only $u \geq 1$ (T1–T3)	12/12; 9/9; pass
SwinIR raises only $u \geq 1$ (ep. 50)	12/12; 9/9; pass
small decoder costs $\leq$ 0.05 dB at $\times 4$+ (D1)	**fail**
no-cell cost decays, $\leq$ 0.03 dB at $\times 4$+ (D2)	**fail** (narrow)
encoder gain $>$ decoder loss (D3)	pass

response analysis of deep super-resolution models and visualization of degradation mismatch," *Nonlinear Theory and Its Applications, IEICE*, vol. 17, no. 4, pp. 1418–1433, 2026.

[11] Wenshuo Wang and Fan Zhang, "Breaking scale anchoring: Frequency representation learning for accurate high-resolution inference from low-resolution training," arXiv:2512.05132v2, 2026.

[12] Bee Lim, Sanghyun Son, Heewon Kim, Seungjun Nah, and Kyoung Mu Lee, "Enhanced deep residual networks for single image super-resolution," in *CVPR Workshops*, 2017.

[13] Eirikur Agustsson and Radu Timofte, "Ntire 2017 challenge on single image super-resolution: Dataset and study," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*, 2017, pp. 126–135.

[14] Yulun Zhang, Yapeng Tian, Yu Kong, Bineng Zhong, and Yun Fu, "Residual dense network for image super-resolution," in *CVPR*, 2018.

[15] Byeonghyun Pak, Jaewon Lee, and Kyong Hwan Jin, "B-spline texture coefficients estimator for screen content image super-resolution," in *CVPR*, 2023.

[16] Jingyun Liang, Jiezhang Cao, Guolei Sun, Kai Zhang, Luc Van Gool, and Radu Timofte, "SwinIR: Image restoration using swin transformer," in *ICCV Workshops*, 2021.

[17] Roman Zeyde, Michael Elad, and Matan Protter, "On single image scale-up using sparse-representations," in *Curves and Surfaces*, pp. 711–730. Springer, 2012.

[18] David R. Martin, Charless Fowlkes, Doron Tal, and Jitendra Malik, "A database of human segmented natural images and its application to evaluating segmentation algorithms and measuring ecological statistics," Tech. Rep. UCB/CSD-01-1133, EECS Department, University of California, Berkeley, 2001.

[19] Jia-Bin Huang, Abhishek Singh, and Narendra Ahuja, "Single image super-resolution from transformed self-exemplars," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2015, pp. 5197–5206.